590442

Course Name: DATABASE MANAGEMENT SYSTEM

Course Outcome

- CO1: Understand the concept of database management systems and Relational database.
- CO2: Identify the various data model used in database design.
- CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
- CO4: Create and populate a RDBMS for a real life application, with constraints and keys using SQL.
- CO5: Select the information from a database by formulating complex queries in SQL.
- CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
- CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
- CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases

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End Term Examination, Even Semester 2024-25
B. TECH. (CSE/EC)- II Year, IV Semester
BCSC1003 : DATABASE MANAGEMENT SYSTEM

Time: 3 Hours **Maximum Marks: 50**

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What is a foreign key constraint? Why are such constraints important? What is referential integrity?	4	CO4	U	F
2	Consider the following schema: Suppliers (sid: integer, sname: string, address: string) Parts (pid: integer, pname: string, color: string) Catalog (sid: integer, pid: integer, cost: real) The key fields are underlined, and the domain of each field is listed after the field name. Therefore, sid is the key for Suppliers, pid is the key for Parts, and sid and pid together form the key for Catalog. The Catalog relation lists the prices charged for parts by Suppliers. Write the following queries in relational algebra. i. Find the names of suppliers who supply some red part. ii. Find the sids of suppliers who supply some red or green part.	4	CO3	E	P

	iii. Find the sids of suppliers who supply some red part or are at 221 Packer Street. iv. 4. Find the sids of suppliers who supply some red part and some green part. (OR) Discuss the following terms with example: i. Generalization ii. Specialization iii. Aggregation				
3	Mention the difference between DELETE, DROP and TRUNCATE Commands in SQL?	4	CO1	R	F
4	Check whether the given schedule S is conflict serializable or not- S : R1(A) , R2(A) , R1(B) , R2(B) , R3(B) , W1(A), W2(B)	4	CO6	E	P
5	A library database stores millions of book records. Users often search books by title or ISBN. How would indexing improve search efficiency in this case? Which type of index would you choose and why?	4	CO7	A	M

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Decompose the following into 4NF EMP ENAME | PNAME | DNAME Dev | X | Sanju Dev | Y | Sainyam Dev | X | Sainyam Dev | Y | Sanju	3	CO6	U	C

7	What are the ACID properties of a transaction? Why is it important? Give an example	3	CO5	A	C
8	What is write-ahead logging? In what ways does it guarantee the transaction durability and consistency?	3	CO7	U	M
9	What are Multi-valued Dependencies? When we can say that a constraint X is multidetermined? (OR) Discuss the concept of Lossless Join and Lossy Decomposition by using an example.	3	CO6	A	M
10	Suppose a relation R(A,B,C) with set of functional dependencies F.D={A->B, B->C, C->D} is decomposed into sub relations R1(A,B), R2(B,C) and R3(A,C). Check that the given dependencies are preserved or not after decomposition.	3	CO6	An	P

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	What is two phase locking? Describe with help of an example. Will two phase locking result in serialisable schedule? Justify your answer with the help of an example.	5	CO7	An	M
12	How can a distributed database system prevent deadlock?	5	CO8	U	C

13	Consider the following schedule- **Transaction T1** | **Transaction T2** R(A) | W(A) | . | R(A) // Dirty Read . | W(A) . | . | . | Commit | Whether a Schedule is Recoverable or Irrecoverable. (OR) Explain the concept of Recoverability of schedules. Discuss any one method/technique for recovering a schedule in the case of any failure with the help of example.	5	CO8	An	P